

Bachelor of Information Technology

Database Systems
BIT1223/BTL1223/BCL1223

Lecture 01 –Introduction to Database



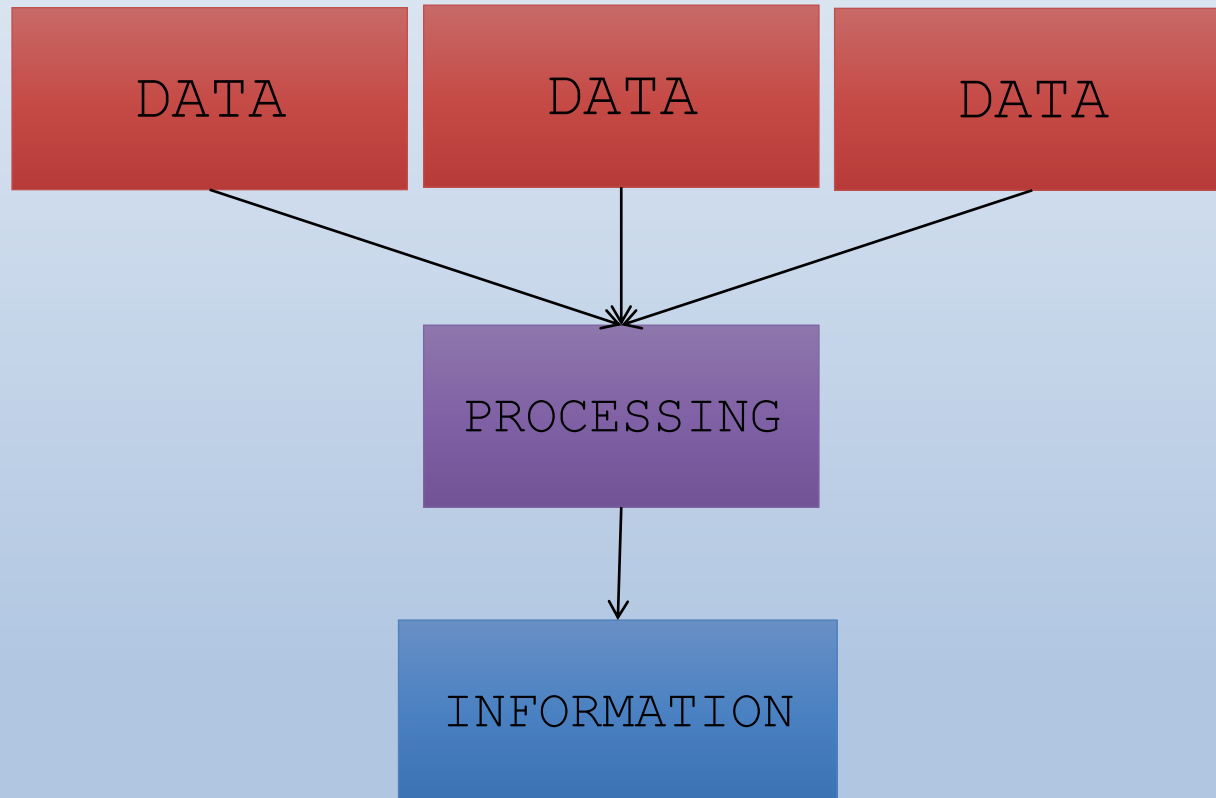
Objectives

- The difference between data and information
- What a database is, what the different types of databases are, and why they are valuable assets for decision making
- The importance of database design
- How modern databases evolved from file systems
- About flaws in file system data management
- What the database system's main components are and how a database system differs from a file system
- The main functions of a database management system (DBMS)

Data vs. Information

- Data:
 - Raw facts; building blocks of information
 - Unprocessed information
- Information:
 - Data processed to reveal meaning
 - Data with context
- Accurate, relevant, and timely information is key to good decision making
- Good decision making is the key to survival in a global environment

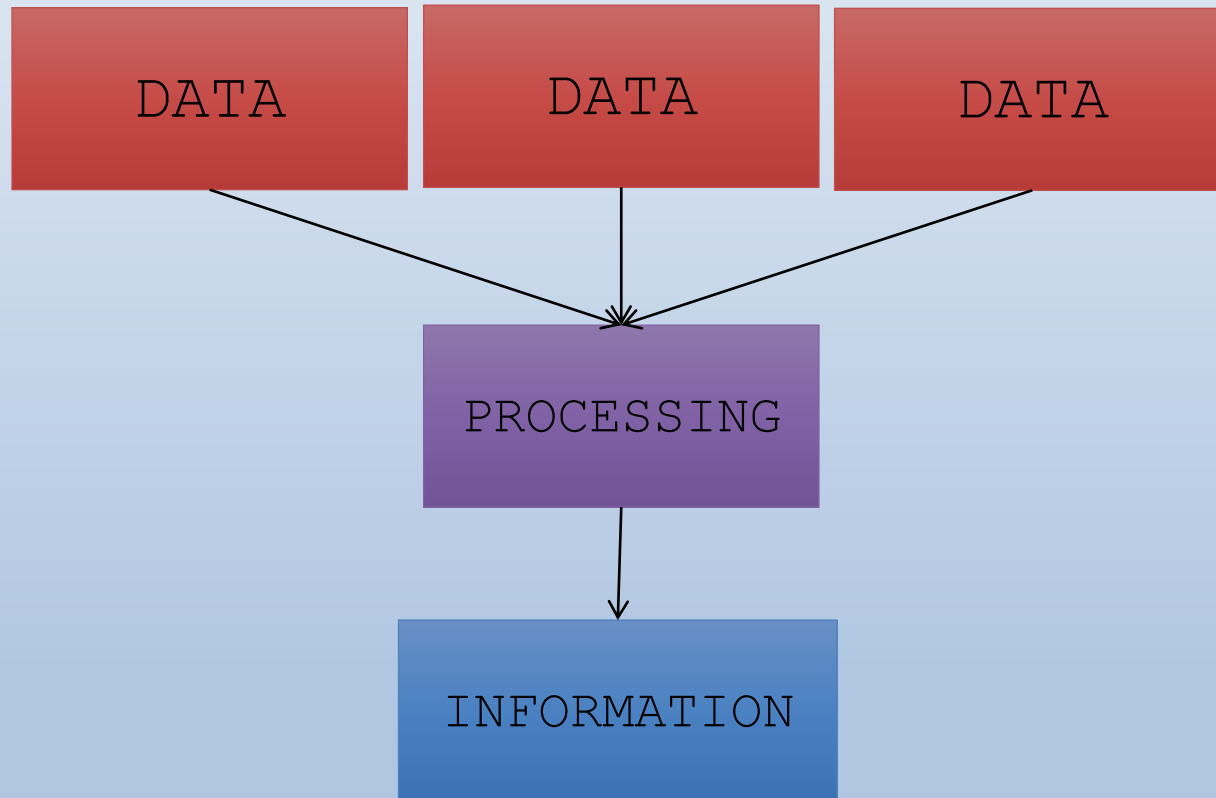
Example



44, 70, 55, 87, 69

Ahmad's score for 6
modules in ICT

Example



Selangor, 1234, Siti,
Jalan 7,Ampang,
0155532111, 43000,
Ahmad

Siti Ahmad
1234 , Jalan 7
43000 Ampang, Selangor
015-5532111

Data vs Information

Parameters	Data	Information
Format	Data is in the form of numbers, letters or a set of characters. Raw data.	Ideas and inferences. Data with context.
Represented in	It can be structured, tabular data, graph, data tree, etc.	Language, ideas, and thought based on the given data
Meaning	Data does not have any specific purpose	It carries meaning that has been assigned by interpreting data
Contains	Unprocessed raw factors	Processed in a meaningful way
Dependency	Data depends upon the sources for collecting data	Information depends upon data
Example	Ticket sales in a band on tour	Sales report by region and venue. It gives information which venue is profitable for the business.

What Data to Store

- Type of Data
 - Raw data
 - Pre-processed
 - Compressed
- Frequency of Data Storage for Sensor Data
 - Trade off between precision and quantity
- Database – shared, integrated computer structure that stores:
 - End user data (raw facts)
 - Metadata (data about data)

Historical Roots: Files and File Systems

- Managing data with file systems is obsolete
 - Understanding file system characteristics makes database design easier to understand
 - Awareness of problems with file systems helps prevent similar problems in DBMS
 - Knowledge of file systems is helpful if you plan to convert an obsolete file system to a DBMS

Historical Roots: Files and File Systems

Manual File systems:

- Collection of file folders kept in file cabinet
- Organization within folders based on data's expected use (ideally logically related)
- System adequate for small amounts of data with few reporting requirements
- Finding and using data in growing collections of file folders became time-consuming and cumbersome

Historical Roots: Files and File Systems

Conversion from manual to computer system:

- Could be technically complex, requiring hiring of data processing (DP) specialists
- Resulted in numerous “home-grown” systems being created
- Initially, computer files were similar in design to manual files

Historical Roots: Files and File Systems

	C_NAME	C_PHONE	C_ADDRESS	C_ZIP	A_NAME	A_PHONE	TP	AMT	REN
▶	Alfred A. Ramas	615-844-2573	218 Fork Rd., Babs, TN	36123	Leah F. Hahn	615-882-1244	T1	\$100.00	05-Apr-2006
	Leona K. Dunne	713-894-1238	Box 12A, Fox, KY	25246	Alex B. Alby	713-228-1249	T1	\$250.00	16-Jun-2006
	Kathy W. Smith	615-894-2285	125 Oak Ln, Babs, TN	36123	Leah F. Hahn	615-882-2144	S2	\$150.00	29-Jan-2007
	Paul F. Olowski	615-894-2180	217 Lee Ln., Babs, TN	36123	Leah F. Hahn	615-882-1244	S1	\$300.00	14-Oct-2006
	Myron Orlando	615-222-1672	Box 111, New, TN	36155	Alex B. Alby	713-228-1249	T1	\$100.00	28-Dec-2006
	Amy B. O'Brian	713-442-3381	387 Troll Dr., Fox, KY	25246	John T. Okon	615-123-5589	T2	\$850.00	22-Sep-2006
	James G. Brown	615-297-1228	21 Tye Rd., Nash, TN	37118	Leah F. Hahn	615-882-1244	S1	\$120.00	25-Mar-2006
	George Williams	615-290-2556	155 Maple, Nash, TN	37119	John T. Okon	615-123-5589	S1	\$250.00	17-Jul-2006
	Anne G. Farriss	713-382-7185	2119 Elm, Crew, KY	25432	Alex B. Alby	713-228-1249	T2	\$100.00	03-Dec-2006
	Olette K. Smith	615-297-3809	2782 Main, Nash, TN	37118	John T. Okon	615-123-5589	S2	\$500.00	14-Mar-2006

C_NAME = Customer name
C_PHONE = Customer phone
C_ADDRESS = Customer address
C_ZIP = Customer zip code

A_NAME = Agent name
A_PHONE = Agent phone
TP = Insurance type
AMT = Insurance policy amount, in thousands of \$
REN = Insurance renewal date

Contents of the CUSTOMER file

Historical Roots: Files and File Systems

TERM	DEFINITION
Data	“Raw” facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data have little meaning unless they have been organized in some logical manner. The smallest piece of data that can be “recognized” by the computer is a single character, such as the letter A, the number 5, or a symbol such as /. A single character requires 1 byte of computer storage.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For example, the fields that constitute a record for a customer named J. D. Rudd might consist of J. D. Rudd’s name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about vendors of ROBCOR Company, or a file might contain the records for the students currently enrolled at Gigantic University.

Basic File Terminology



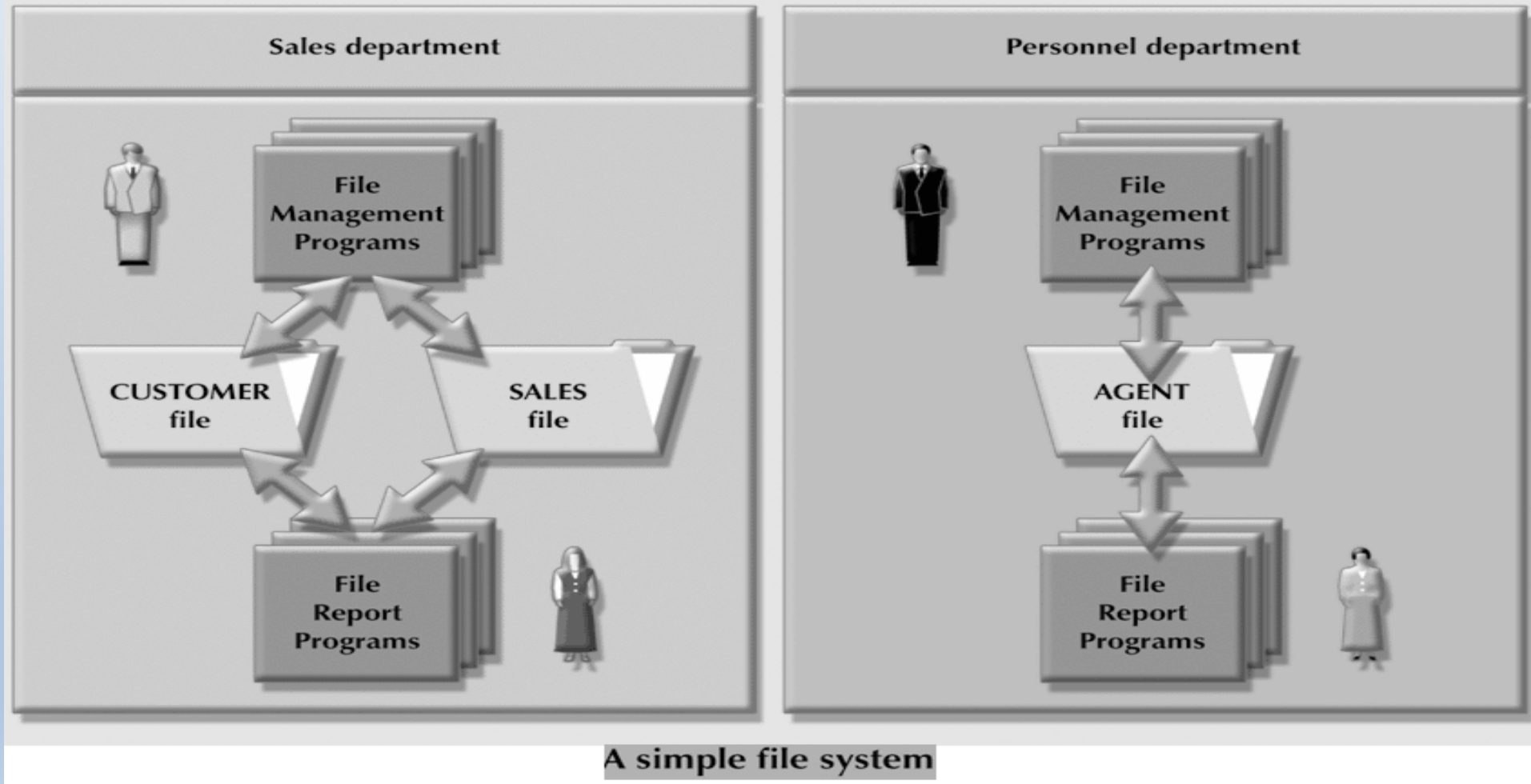
Historical Roots: Files and File Systems

- DP specialist wrote programs for reports:
 - Monthly summaries of types and amounts of insurance sold by agents
 - Monthly reports about which customers should be contacted for renewal
 - Reports that analyzed ratios of insurance types sold by agent
 - Customer contact letters summarizing coverage

Historical Roots: Files and File Systems

- Other departments requested databases be written for them
 - SALES database created for sales department
 - AGENT database created for personnel department
- As number of databases increased, small file system evolved
- Each file used its own application programs
- Each file was owned by individual or department who commissioned its creation

Historical Roots: Files and File Systems



Example of Early Database Design (continued)

- As system grew, demand for DP's programming skills grew
- Additional programmers hired
- DP specialist evolved into DP manager, supervising a DP department
- Primary activity of department (and DP manager) remained programming

Problems with File System Data Management

- Every task requires extensive programming in a third-generation language (3GL)
 - Programmer must specify task and how it must be done
- Modern databases use fourth-generation languages (4GL)
 - Allow users to specify what must be done without specifying how it is to be done
- Example: DO Loop VS. Select Statement

Problems with File System Data Management

- Time-consuming, high-level activity
- As number of files expands, system administration becomes difficult
- Making changes in existing file structure is difficult
- File structure changes require modifications in all programs that use data in that file
- Modifications are likely to produce errors, requiring additional time to “debug” the program
- Security features hard to program and therefore often omitted

Structural and Data Dependence

- Structural dependence
 - Access to a file depends on its structure
- Data dependence
 - Changes in the data storage characteristics without affecting the application program's ability to access the data
 - Logical data format
 - How the human being views the data
 - Physical data format
 - How the computer “sees” the data

Field Definitions and Naming Conventions

- Flexible record definition anticipates reporting requirements by breaking up fields into their component parts
- Example:
Customer Last Name ... Cus-LName

Data Redundancy

- Data redundancy results in data inconsistency
 - Different and conflicting versions of the same data appear in different places
- Errors more likely to occur when complex entries are made in several different files and/or recur frequently in one or more files
- Data anomalies develop when required changes in redundant data are not made successfully

Data Redundancy

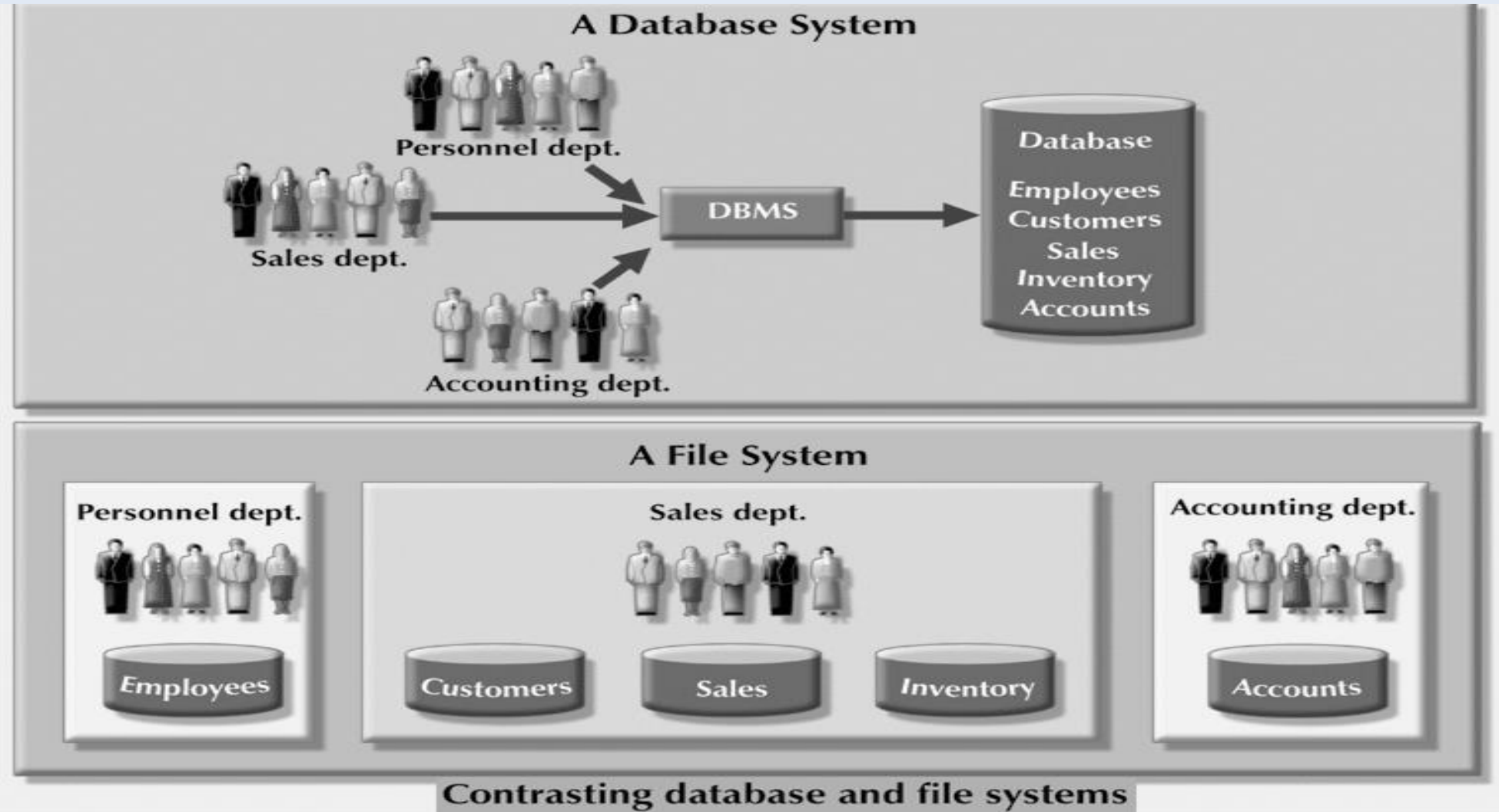
Types of data anomalies:

- Update anomalies
 - Occur when changes must be made to existing records
- Insertion anomalies
 - Occur when entering new records
- Deletion anomalies
 - Occur when deleting records

Database Systems

- Problems inherent in file systems make using a database system desirable
- File system
 - Many separate and unrelated files
- Database
 - Logically related data stored in a single logical data repository

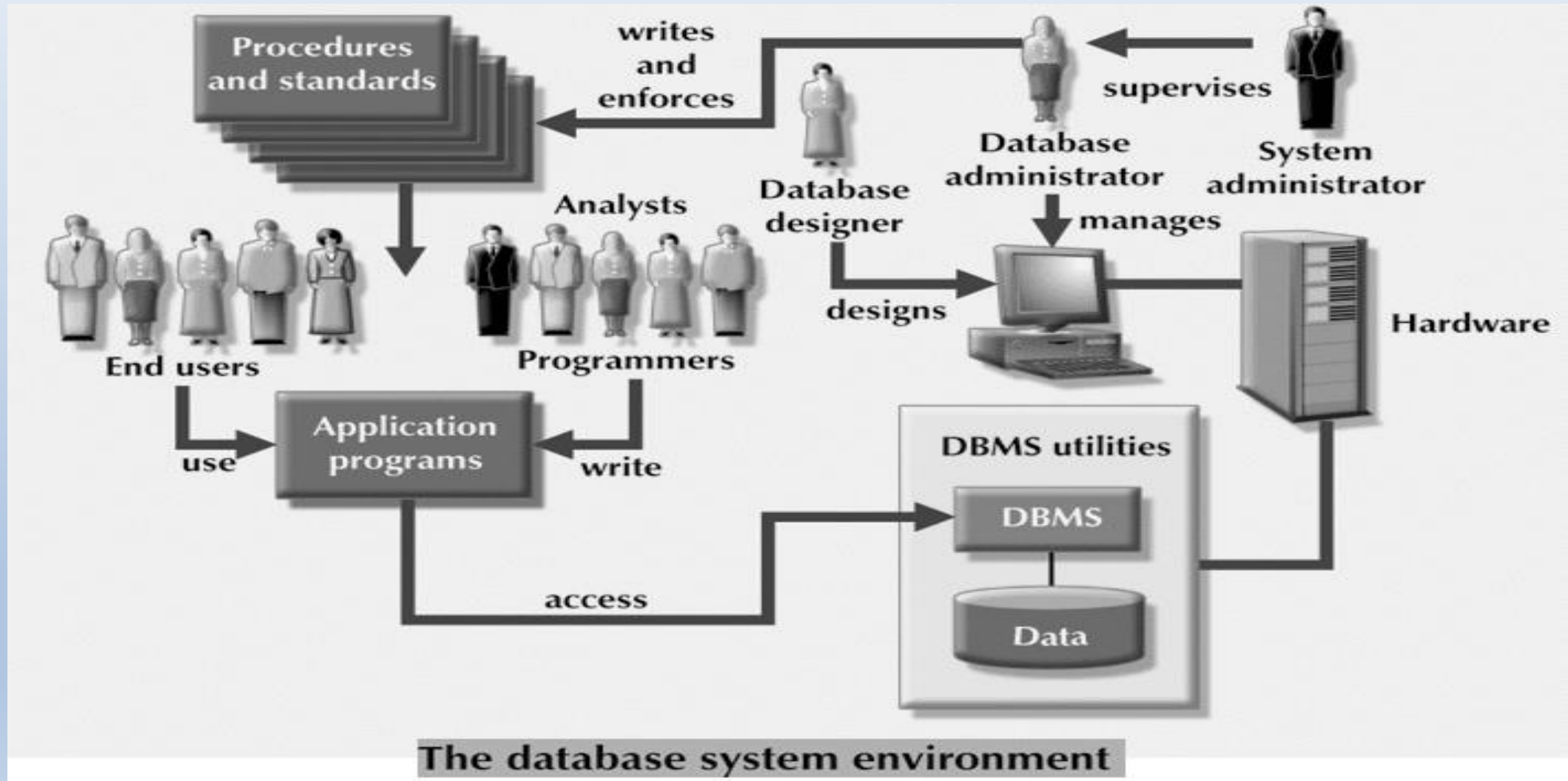
Database Systems



The Database System Environment

- Database system is composed of five main parts:
 - Hardware
 - Software
 - Operating system software
 - DBMS software
 - Application programs and utility software
 - People
 - Procedures
 - Data

The Database System Environment



Introduction to Database Systems

- DBMS (database management system) :
 - Collection of programs that manages database structure and controls access to data
 - Possible to share data among multiple applications or users
 - Makes data management more efficient and effective

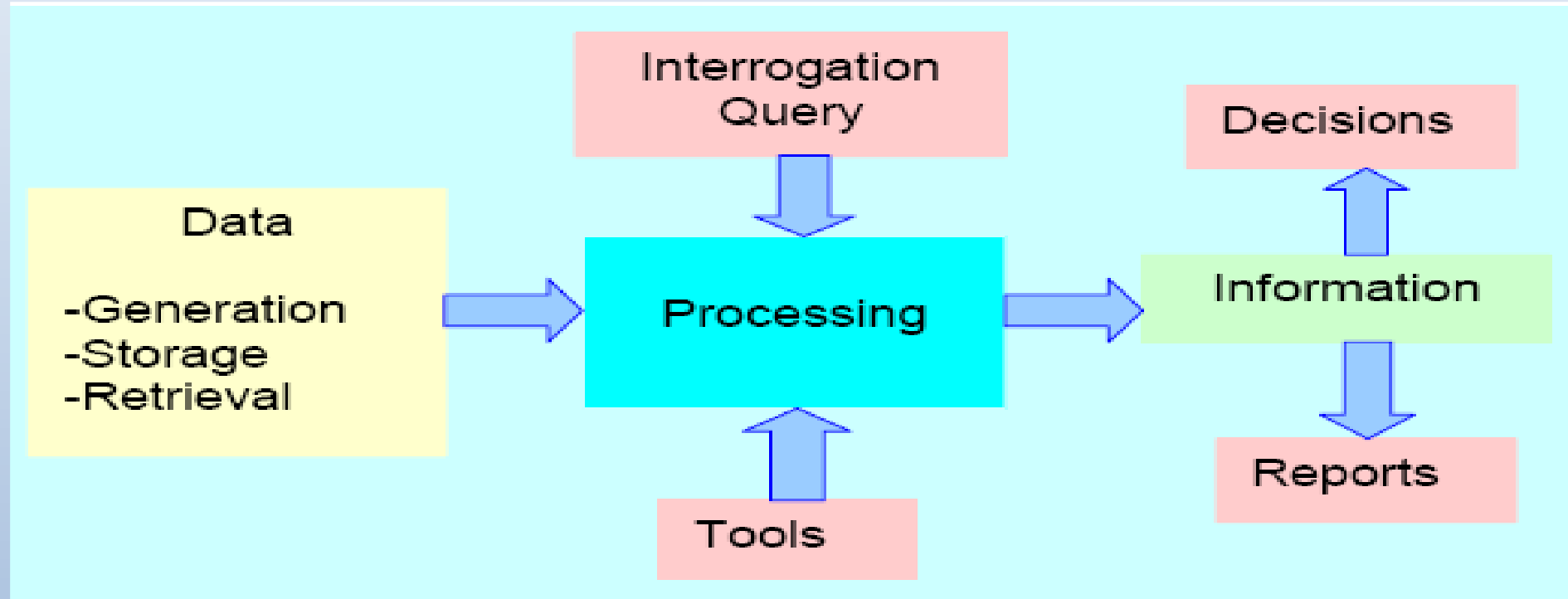
School Table

ID	Name
S001	University of Technology
S002	University of Applied Science

Student Table

School ID	ID	Name	DOB
S001	UT-1000	Tommy	05/06/1995
S001	UT-1000	Better	16/04/1995
S002	UAS-1000	Linda	02/09/1995
S002	UAS-1000	Jonathan	22/06/1995

Basic components of a database system

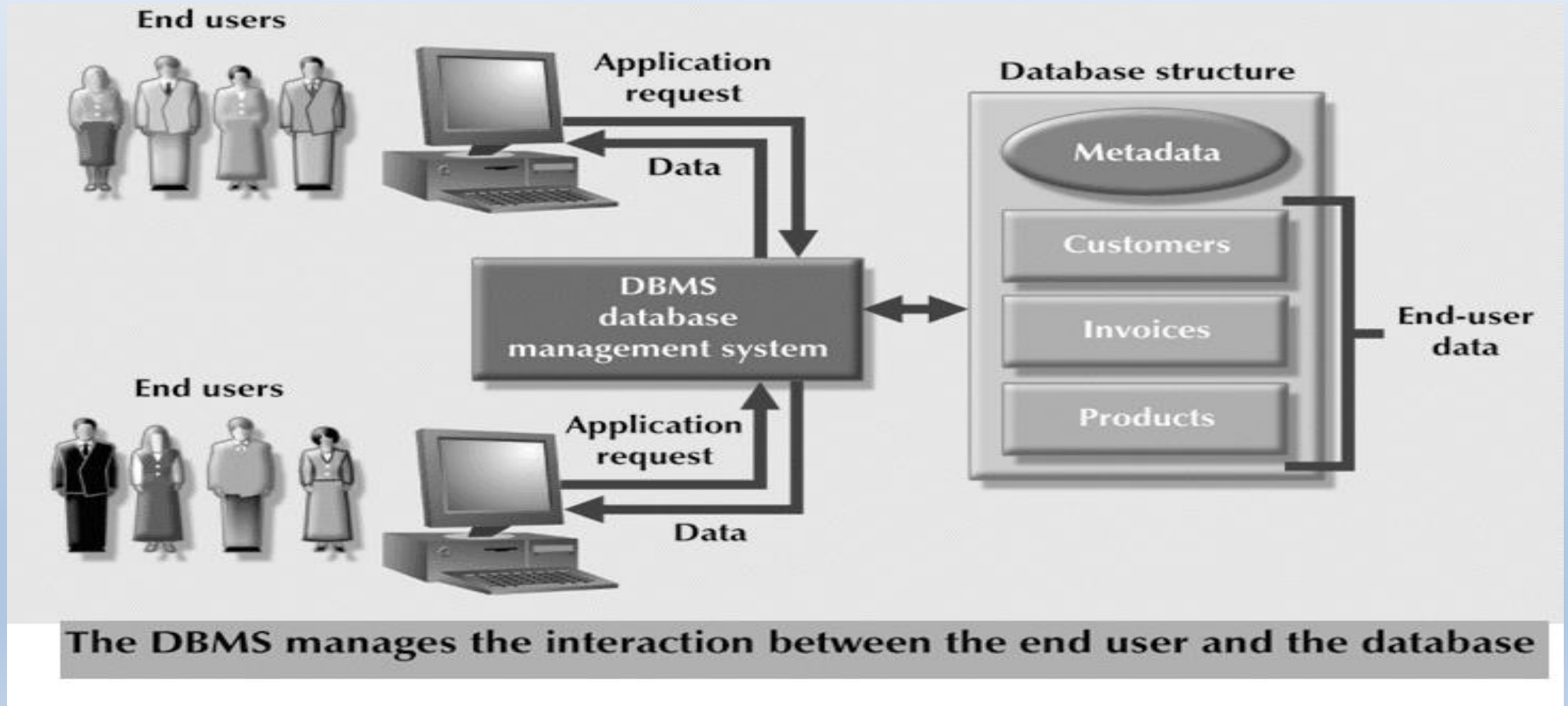


In order to convert data into useful information a set of software tools are need, SQL, Form, etc..

Role and Advantages of the DBMS

- End users have better access to more and better-managed data
 - Promotes integrated view of organization's operations
 - Probability of data inconsistency is greatly reduced
 - Possible to produce quick answers to ad hoc queries

Role and Advantages of the DBMS



Why Database Design is Important

- Defines the database's expected use
- Different approach needed for different types of databases
- Avoid redundant data
- Poorly designed database generates errors
 - leads to bad decisions
 - can lead to failure of organization

DBMS Functions

- DBMS performs functions that guarantee integrity and consistency of data
 1. Data dictionary management
 - defines data elements and their relationships
 2. Data storage management
 - stores data and related data entry forms, report definitions, etc.
 3. Data transformation and presentation
 - translates logical requests into commands to physically locate and retrieve the requested data
 4. Security management
 - enforces user security and data privacy within database

DBMS Functions

5. Multiuser access control

- uses sophisticated algorithms to ensure multiple users can access the database concurrently without compromising the integrity of the database

6. Backup and recovery management

- provides backup and data recovery procedures

7. Data integrity management

- promotes and enforces integrity rules

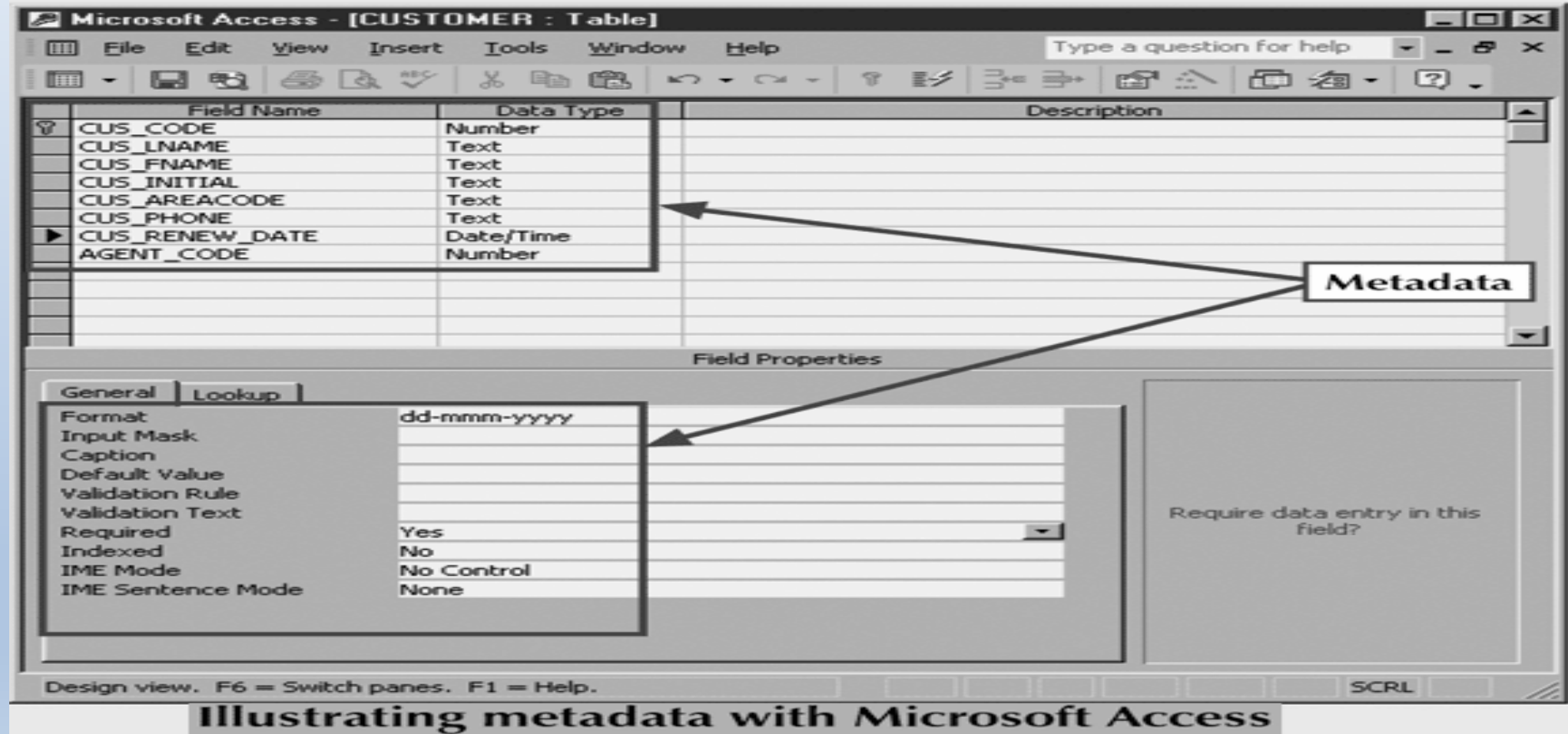
8. Database access languages and application programming interfaces

- provide data access through a query language

9. Database communication interfaces

- allow database to accept end-user requests via multiple, different network environments

DBMS Functions



File based Vs Database

File based	Database
1. The data is distributed in different files and cannot be shared	1. The data is stored at one place and can be shared easily
2. The data integrity checks are difficult to apply on files	2. It provide constraints for data integrity
3. It provide poor security as data is widely spread	3. It provide data security
4. It is less complex system	4. It is very complex system
5. The cost is very less than dbms	5. It is costly system and more chance of failure

Summary

- Data are raw facts. Information is the result of processing data to reveal its meaning.
- To implement and manage a database, use a DBMS.
- Database design defines the database structure.
- A well-designed database facilitates data management and generates accurate and valuable information.
- A poorly designed database can lead to bad decision making, and bad decision making can lead to the failure of an organization.

Summary (continued)

- Databases were preceded by file systems.
- Limitations of file system data management:
 - requires extensive programming
 - system administration complex and difficult
 - making changes to existing structures is difficult
 - security features are likely to be inadequate
 - independent files tend to contain redundant data
- DBMS's were developed to address file systems' inherent weaknesses

**THANK YOU
ANY QUESTIONS?**